

$$u_{32} = (-1)^5 \cdot \begin{vmatrix} 1 & 7 \\ 1 & 0 \end{vmatrix} = 7$$

$$u_{33} = (-1)^6 \cdot \begin{vmatrix} 1 & 3 \\ 1 & 7 \end{vmatrix} = -2$$

$$A_c = \begin{bmatrix} 3 & 3 & 2 \\ -7 & 3 & 2 \\ -1 & 1 & -2 \end{bmatrix} = u c^T = \begin{bmatrix} 3 & -7 & -1 \\ -3 & 3 & 1 \\ 2 & -2 & -3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{-4} \cdot \begin{bmatrix} 3 & -7 & -1 \\ -3 & 3 & 1 \\ 2 & -2 & -2 \end{bmatrix} = \begin{bmatrix} -3/4 & 7/4 & 1/4 \\ 3/4 & -3/4 & -1/4 \\ -1/2 & 1/2 & 1/2 \end{bmatrix}$$

$$\text{Sistema} = \begin{cases} 7x + 3y + z = 10 \\ x + y + 0z = 1 \\ 20x + 2y + 3z = 13 \end{cases}$$

Inverso da matriz do sistema

$$A = \begin{bmatrix} 7 & 3 & 1 \\ 1 & 1 & 0 \\ 20 & 2 & 3 \end{bmatrix}$$

$$\text{Det } A = (-1)^3 \cdot \begin{vmatrix} 3 & 1 \\ 2 & 3 \end{vmatrix} + (-1)^2 \cdot \begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix} =$$

$$\Rightarrow (-1) \cdot 7 + 3 = -4$$

$$a_{11} = (-1)^2 \cdot \begin{vmatrix} 1 & 0 \\ 2 & 3 \end{vmatrix} = 3$$

$$a_{12} = (-1)^3 \cdot \begin{vmatrix} 1 & 0 \\ 0 & 3 \end{vmatrix} = -3$$

$$a_{13} = (-1)^1 \cdot \begin{vmatrix} 1 & 1 \\ 0 & 2 \end{vmatrix} = -2$$

$$a_{21} = (-1)^3 \cdot \begin{vmatrix} 3 & 1 \\ 2 & 3 \end{vmatrix} = -7$$

$$a_{22} = (-1)^4 \cdot \begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix} = 3$$

$$a_{23} = (-1)^5 \cdot \begin{vmatrix} 1 & 3 \\ 0 & 2 \end{vmatrix} = -2$$

$$a_{31} = (-1)^4 \cdot \begin{vmatrix} 3 & 1 \\ 1 & 0 \end{vmatrix} = -1$$